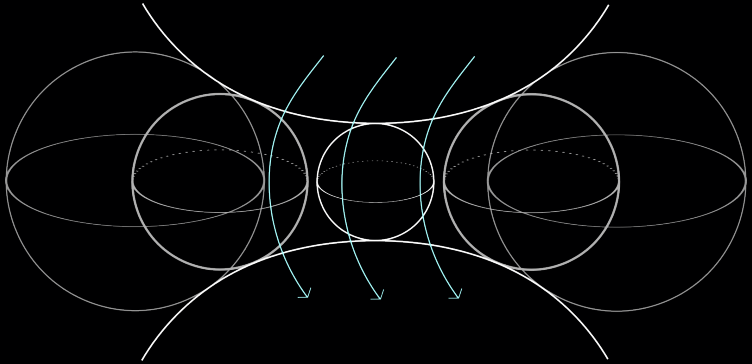


Spaces of Positive Scalar Curvature Metrics on Non-Orientable Manifolds

Milan Zerbin Madrid, 30.04.25



The World of Positive Scalar Curvature

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n.

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature is a function $\mathrm{scal}_g \colon M \to \mathbb{R}$ satisfying:

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature is a function $\mathrm{scal}_g \colon M \to \mathbb{R}$ satisfying:

- Scaling: $\mathrm{scal}_{\lambda g}(x) = \frac{1}{\lambda}\mathrm{scal}_g(x)$ for all $x \in M$ and $\lambda \in \mathbb{R}$.

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature is a function $\mathrm{scal}_g \colon M \to \mathbb{R}$ satisfying:

- Scaling: $\mathrm{scal}_{\lambda g}(x) = \frac{1}{\lambda}\mathrm{scal}_g(x)$ for all $x \in M$ and $\lambda \in \mathbb{R}$.
- Additivity: $\mathrm{scal}_{g \oplus g'}(x, x') = \mathrm{scal}_g(x) + \mathrm{scal}_{g'}(x')$ for any $(x, x') \in M \times M'$ and g' on M'.

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature is a function $\mathrm{scal}_g \colon M \to \mathbb{R}$ satisfying:

- Scaling: $\mathrm{scal}_{\lambda g}(x) = \frac{1}{\lambda}\mathrm{scal}_g(x)$ for all $x \in M$ and $\lambda \in \mathbb{R}$.
- Additivity: $\mathrm{scal}_{g \oplus g'}(x, x') = \mathrm{scal}_g(x) + \mathrm{scal}_{g'}(x')$ for any $(x, x') \in M \times M'$ and g' on M'.
- Volume Comparison: $\mathrm{scal}_g(x) < \mathrm{scal}_{g'}(x')$ implies $\mathrm{vol}_g(B_\varepsilon(x)) > \mathrm{vol}_{g'}(B_\varepsilon(x'))$

Scalar Curvature

Let (M, g) be a Riemannian manifold of dimension n. The scalar curvature is a function $\mathrm{scal}_g \colon M \to \mathbb{R}$ satisfying:

- Scaling: $\mathrm{scal}_{\lambda g}(x) = \frac{1}{\lambda}\mathrm{scal}_g(x)$ for all $x \in M$ and $\lambda \in \mathbb{R}$.
- Additivity: $\mathrm{scal}_{g \oplus g'}(x, x') = \mathrm{scal}_g(x) + \mathrm{scal}_{g'}(x')$ for any $(x, x') \in M \times M'$ and g' on M'.
- Volume Comparison: $\mathrm{scal}_g(x) < \mathrm{scal}_{g'}(x')$ implies $\mathrm{vol}_g(B_\varepsilon(x)) > \mathrm{vol}_{g'}(B_\varepsilon(x'))$

It is up to normalization uniquely determined by those properties. [Gromov 2021]

Examples

- The sphere $S^n \subset \mathbb{R}^{n+1}$ has constant positive scalar curvature when $n > 1$.

Examples

- The sphere $S^n \subset \mathbb{R}^{n+1}$ has constant positive scalar curvature when $n > 1$.
- Euclidean space $(\mathbb{R}^n, g_{\text{euklid}})$ has vanishing scalar curvature.

Examples

- The sphere $S^n \subset \mathbb{R}^{n+1}$ has constant positive scalar curvature when $n > 1$.
- Euclidean space $(\mathbb{R}^n, g_{\text{euklid}})$ has vanishing scalar curvature.
- Any onedimensional Riemannian manifold has vanishing scalar curvature.

Examples

- The sphere $S^n \subset \mathbb{R}^{n+1}$ has constant positive scalar curvature when $n > 1$.
- Euclidean space $(\mathbb{R}^n, g_{\text{euklid}})$ has vanishing scalar curvature.
- Any onedimensional Riemannian manifold has vanishing scalar curvature.
- The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ can be given a metric of scalar curvature zero.

Examples

- The sphere $S^n \subset \mathbb{R}^{n+1}$ has constant positive scalar curvature when $n > 1$.
- Euclidean space $(\mathbb{R}^n, g_{\text{euklid}})$ has vanishing scalar curvature.
- Any onedimensional Riemannian manifold has vanishing scalar curvature.
- The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$ can be given a metric of scalar curvature zero.
- The projective space $\mathbb{R}\mathrm{P}^n$ can be given a metric of positive scalar curvature when $n > 1$.

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K.

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K. The Gauss–Bonnet-theorem says

$$\int_\Sigma K \,\mathrm{d}A = 2\pi\, \chi(M)$$

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K. The Gauss–Bonnet-theorem says

$$\int_{\Sigma} K \, \mathrm{d}A = 2\pi \, \chi(M)$$

So the only surfaces that admits a metric of positive scalar curvature are

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K. The Gauss–Bonnet-theorem says

$$\int_\Sigma K \, \mathrm{d}A = 2\pi \, \chi(M)$$

So the only surfaces that admits a metric of positive scalar curvature are S^2 and $\mathbb{R}\mathrm{P}^2$.

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K. The Gauss–Bonnet-theorem says

$$\int_\Sigma K \,\mathrm{d}A = 2\pi\, \chi(M)$$

So the only surfaces that admits a metric of positive scalar curvature are S^2 and $\mathbb{R}\mathrm{P}^2$.
$\rightarrow$ Existence settled, how about uniqueness?

Dimension 2

Scalar curvature of a surface (Σ, g) is just twice the Gaussian curvature K. The Gauss–Bonnet-theorem says

$$\int_\Sigma K \,\mathrm{d}A = 2\pi\, \chi(M)$$

So the only surfaces that admits a metric of positive scalar curvature are S^2 and $\mathbb{R}\mathrm{P}^2$.
$\rightarrow$ Existence settled, how about uniqueness?
$\rightarrow$ $\mathcal{R}^+(S^2)$ and $\mathcal{R}^+(\mathbb{R}\mathrm{P}^2)$ even contractible [Rosenberg-Stolz 2001]

Dimension ≥ 3

Notation:

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

- M always admits a metric of negative scalar curvature. [Aubin 1970]

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

- M always admits a metric of negative scalar curvature. [Aubin 1970]
- The space $\mathcal{R}^-(M)$ is always contractible. [Lohkamp 1992]

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

- M always admits a metric of negative scalar curvature. [Aubin 1970]
- The space $\mathcal{R}^-(M)$ is always contractible. [Lohkamp 1992]
- If M' has PSC, then so does $M' \times M$ for any M.

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

- M always admits a metric of negative scalar curvature. [Aubin 1970]
- The space $\mathcal{R}^-(M)$ is always contractible. [Lohkamp 1992]
- If M' has PSC, then so does $M' \times M$ for any M.
- If M and M' have PSC, then so does $M \# M'$.

Dimension ≥ 3

Notation: A manifold M is called PSC-manifold, if it admits a metric of positive scalar curvature.

- M always admits a metric of negative scalar curvature. [Aubin 1970]
- The space $\mathcal{R}^-(M)$ is always contractible. [Lohkamp 1992]
- If M' has PSC, then so does $M' \times M$ for any M.
- If M and M' have PSC, then so does $M \# M'$.

Existence in high dimensions

If M is obtained from M' by a surgery of codimension at least 3, then $\mathcal{R}^+(M) \neq \varnothing \Rightarrow \mathcal{R}^+(M') \neq \varnothing$.

Existence in high dimensions

If M is obtained from M' by a surgery of codimension at least 3, then $\mathcal{R}^+(M) \neq \varnothing \Rightarrow \mathcal{R}^+(M') \neq \varnothing$. Used to prove:

Theorem (Gromov–Lawson, 1980)

Let M be a simply connected manifold of dimension $n \geq 5$. If M is not spinnable, M is a PSC-manifold.

Existence in high dimensions

If M is obtained from M' by a surgery of codimension at least 3, then $\mathcal{R}^+(M) \neq \varnothing \Rightarrow \mathcal{R}^+(M') \neq \varnothing$. Used to prove:

Theorem (Gromov–Lawson, 1980)

Let M be a simply connected manifold of dimension $n \geq 5$. If M is not spinnable, M is a PSC-manifold.

$\rightarrow$ Spinnable case by Stolz in 1992: Vanishing of $\hat{\mathfrak{a}}(M)$ neccesary and sufficient.

Uniqueness

Theorem (Ebert–Wiemeler, 2022)

Let M be a simply connected spin manifold of dimension $n \geq 5$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^n)$$

whenever M admits a PSC-metric.

Uniqueness

Theorem (Ebert–Wiemeler, 2022)

Let M be a simply connected spin manifold of dimension $n \geq 5$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^n)$$

whenever M admits a PSC-metric.

$\rightarrow$ Simply connected, nonspinnable case similarly handled.

Uniqueness

Theorem (Ebert–Wiemeler, 2022)

Let M be a simply connected spin manifold of dimension $n \geq 5$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^n)$$

whenever M admits a PSC-metric.

$\rightarrow$ Simply connected, nonspinnable case similarly handled.
$\rightarrow$ $\mathcal{R}^+(S^n)$ has rich topology for $n \geq 6$ [Ebert–Randal-Williams 2019]

Method

Theorem (Ebert–Wiemeler, 2022)

Let M and M' be n-manifolds of tangential 2-type (B, θ) and $x \in \Omega_n^\theta$ good, such that $[M] \equiv [M'] + x$ in Ω_n^θ. Then if $n \geq 5$

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(M').$$

Method

Theorem (Ebert–Wiemeler, 2022)

Let M and M' be n-manifolds of tangential 2-type (B, θ) and $x \in \Omega_n^\theta$ good, such that $[M] \equiv [M'] + x$ in Ω_n^θ. Then if $n \geq 5$

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(M').$$

$\rightarrow$ There are six tangential 2-types. For each one, obtain a theorem.

Method

Theorem (Ebert–Wiemeler, 2022)

Let M and M' be n-manifolds of tangential 2-type (B, θ) and $x \in \Omega_n^\theta$ good, such that $[M] \equiv [M'] + x$ in Ω_n^θ. Then if $n \geq 5$

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(M').$$

$\rightarrow$ There are six tangential 2-types. For each one, obtain a theorem.
$\rightarrow \theta = \mathrm{BSpin} \rightarrow \mathrm{BO}$ gives the result on the previous slide

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Prove that all those generators are good.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Prove that all those generators are good.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ).

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Prove that all those generators are good.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ).

$\rightarrow$ In other words, there is only one possible homotopy type of $\mathcal{R}^+(M)$ for manifolds of this tangential 2-type in dimension n.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Prove that all those generators are good. ← What if they aren't?
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ).

→ In other words, there is only one possible homotopy type of $\mathcal{R}^+(M)$ for manifolds of this tangential 2-type in dimension n.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Find a (hopefully large) subset of of good generators.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ) and $M \equiv W$ in $\Omega_n^\theta / \mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.

Step-by-Step

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ) and $M \equiv W$ in $\Omega_n^\theta / \mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.

$\rightarrow$ Notice, that $\left|\Omega_n^\theta / \mathcal{A}_n\right|$ is an upper bound for the amount of different homotopy types of $\mathcal{R}^+(M)$ in dimension n.

Step-by-Step in our Case

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type (B, θ).

1. Compute a set of generators of Ω_n^θ.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ) and $M \equiv W$ in $\Omega_n^\theta / \mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.

Step-by-Step in our Case

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type $(\mathrm{BO}, \mathrm{id}_{\mathrm{BO}})$.

1. Compute a set of generators of Ω_n^θ.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ) and $M \equiv W$ in $\Omega_n^\theta / \mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.

Step-by-Step in our Case

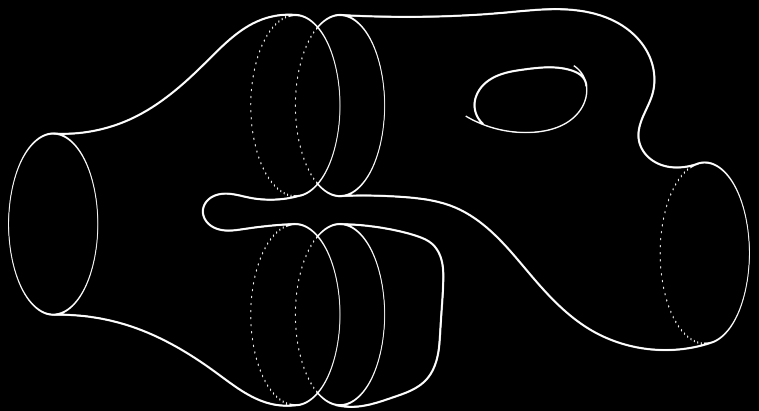
Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type $(\mathrm{BO}, \mathrm{id}_{\mathrm{BO}})$.

1. Compute a set of generators of $\mathfrak{N}_n$.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type (B, θ) and $M \equiv W$ in $\Omega_n^\theta / \mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.

Step-by-Step in our Case

Fix a dimension $n \geq 5$. Let W be an n-dimensional manifold of tangential 2-type $(\mathrm{BO}, \mathrm{id}_{\mathrm{BO}})$.

1. Compute a set of generators of $\mathfrak{N}_n$.
2. Find a (hopefully large) subset of of good generators.
3. Obtain $\mathcal{R}^+(M) \simeq \mathcal{R}^+(W)$ for any M with tangential 2-type $(\mathrm{BO}, \mathrm{id}_{\mathrm{BO}})$ and $M \equiv W$ in $\mathfrak{N}_n/\mathcal{A}_n$, where $\mathcal{A}_n$ is the subgroup generated by the good generators.



Calculations in the Unoriented Bordism Groups

Bordism

Definition

Call two closed n-dimensional manifolds M, M' bordant, if there is a manifold W of dimension $n+1$ with boundary $\partial W = M \sqcup M'$.

Bordism

Definition

Call two closed n-dimensional manifolds M, M' bordant, if there is a manifold W of dimension $n+1$ with boundary $\partial W = M \sqcup M'$.

$\rightarrow$ Defines an equivalence relation on the set of all closed n-dimensional manifolds.

Bordism

Definition

Call two closed n-dimensional manifolds M, M' bordant, if there is a manifold W of dimension $n+1$ with boundary $\partial W = M \sqcup M'$.

$\rightarrow$ Defines an equivalence relation on the set of all closed n-dimensional manifolds.
$\rightarrow$ The set of equivalence classes forms an abelian group $\mathfrak{N}_n$ with disjoint union $\sqcup$ as sum.

Ring Structure

Definition

The (unoriented) bordism ring is the unital, commutative, graded ring

$$\mathfrak{N}_* = \left(\bigcup_{n \in \mathbb{N}} \mathfrak{N}_n, \sqcup, \times, [\varnothing], [\{*\}] \right)$$

Ring Structure

Definition

The (unoriented) bordism ring is the unital, commutative, graded ring

$$\mathfrak{N}_* = \left(\bigcup_{n \in \mathbb{N}} \mathfrak{N}_n, \sqcup, \times, [\varnothing], [\{*\}] \right)$$

$\rightarrow$ Surprisingly, $\mathfrak{N}_n$ is finite for each n.

Ring Structure

Definition

The (unoriented) bordism ring is the unital, commutative, graded ring

$$\mathfrak{N}_* = \left(\bigcup_{n \in \mathbb{N}} \mathfrak{N}_n, \sqcup, \times, [\varnothing], [\{*\}] \right)$$

$\rightarrow$ Surprisingly, $\mathfrak{N}_n$ is finite for each n.
$\rightarrow$ Furthermore, the entire ring is well understood...

Thoms work

Theorem (Thom–Dold, 1954-1956)

The unoriented bordism ring is isomorphic to a polynomial algebra over $\mathbb{Z}_2$, with precicely one generator in each degree $n \neq 2^k - 1$. Generators are represented by Dold-manifolds $P(m, n)$.

Thoms work

Theorem (Thom–Dold, 1954-1956)

The unoriented bordism ring is isomorphic to a polynomial algebra over $\mathbb{Z}_2$, with precicely one generator in each degree $n \neq 2^k - 1$. Generators are represented by Dold-manifolds $P(m, n)$.

$\rightarrow$ Define $P(m, n) = S^m \times \mathbb{C}\mathrm{P}^n/_\sim$ where $(x, z) \sim (-x, \bar{z})$.

Thoms work

Theorem (Thom–Dold, 1954-1956)

The unoriented bordism ring is isomorphic to a polynomial algebra over $\mathbb{Z}_2$, with precicely one generator in each degree $n \neq 2^k - 1$. Generators are represented by Dold-manifolds $P(m, n)$.

$\rightarrow$ Define $P(m, n) = S^m \times \mathbb{C}\mathrm{P}^n/_\sim$ where $(x, z) \sim (-x, \bar{z})$.
$\rightarrow$ Notice $P(m, 0) \cong \mathbb{R}\mathrm{P}^m$.

Thoms work more precicely

- In even dimension $n = 2m$, we have a multiplicative generator given by $[P(2m, 0)] = [\mathbb{R}\mathrm{P}^{2m}]$.

Thoms work more precicely

- In even dimension $n = 2m$, we have a multiplicative generator given by $[P(2m, 0)] = [\mathbb{R}\mathrm{P}^{2m}]$.
- In odd dimension $n = 2^r(2s+1) - 1$, we have a multiplicative generator given by $[P(2^r - 1, s2^r)]$ whenever $n \neq 2^k - 1$.

Thoms work more precicely

- In even dimension $n = 2m$, we have a multiplicative generator given by $[P(2m, 0)] = [\mathbb{R}\mathrm{P}^{2m}]$.
- In odd dimension $n = 2^r(2s+1) - 1$, we have a multiplicative generator given by $[P(2^r - 1, s2^r)]$ whenever $n \neq 2^k - 1$.

$\rightarrow$ Additive generators of $\mathfrak{N}_n$ correspond to partitions of n that contain no element of form $2^k - 1$.

Additive Generators

An additive generator of $\mathfrak{N}_n$ has the general form

$$\left[\prod_{i\in I}(\mathbb{R}\mathrm{P}^{2i})^{m_i}\times\prod_{j\in J}P(2^{r_j}-1,s_j2^{r_j})\right]$$

where $\sum_{i\in I}(2m_ii)+\sum_{j\in J}(2^{r_j}(2s_j+1)-1)=n$, and $I,J\subset\mathbb{N}$.

Additive Generators

An additive generator of $\mathfrak{N}_n$ has the general form

$$\left[\prod_{i\in I}(\mathbb{R}\mathrm{P}^{2i})^{m_i} \times \prod_{j\in J} P(2^{r_j}-1, s_j 2^{r_j})\right]$$

where $\sum_{i\in I}(2m_i i) + \sum_{j\in J}(2^{r_j}(2s_j+1)-1) = n$, and $I, J \subset \mathbb{N}$.

$\rightarrow$ Now: Which of those are good?

Good Generators I

Lemma (Ebert–Wiemeler, 2022)

Let $M \to M'$ be a smooth fiber bundle with fiber $\mathbb{K}\mathrm{P}^{n+m-1}$. Then $[M]$ is good if either ...

- $\mathbb{K} = \mathbb{R}$, $n, m \geq 3$ with structure group $O(m) \times O(n)/\{\pm 1\}$
- $\mathbb{K} = \mathbb{C}$, $n, m \geq 2$ with structure group $U(n) \times U(m)$.
- $\mathbb{K} = \mathbb{C}$, $n + m - 1 \geq 3$ with structure group $\mathbb{Z}_2$.
- $\mathbb{K} = \mathbb{H}$, $n, m \geq 1$ with structure group $\mathrm{Sp}(n) \times \mathrm{Sp}(\mathrm{m})/\{\pm 1\}$

Good Generators I

Lemma (Ebert–Wiemeler, 2022)

Let $M \to M'$ be a smooth fiber bundle with fiber $\mathbb{K}\mathrm{P}^{n+m-1}$. Then $[M]$ is good if either ...

- $\mathbb{K} = \mathbb{R}$, $n, m \geq 3$ with structure group $O(m) \times O(n)/\{\pm 1\}$
- $\mathbb{K} = \mathbb{C}$, $n, m \geq 2$ with structure group $U(n) \times U(m)$.
- $\mathbb{K} = \mathbb{C}$, $n + m - 1 \geq 3$ with structure group $\mathbb{Z}_2$.
- $\mathbb{K} = \mathbb{H}$, $n, m \geq 1$ with structure group $\mathrm{Sp}(n) \times \mathrm{Sp}(\mathrm{m})/\{\pm 1\}$

$\to$ Remember Rokhlin's Trick: $[\mathbb{R}\mathrm{P}^n \times \mathbb{R}\mathrm{P}^n] = [\mathbb{C}\mathrm{P}^n]$, $[\mathbb{C}\mathrm{P}^n \times \mathbb{C}\mathrm{P}^n] = [\mathbb{H}\mathrm{P}^n]$.

Good Generators II

Repeatedly applying the above yields, that all (additive) generators are good, except maybe those of form

$$(\mathbb{R}\mathrm{P}^2)^i \times (\mathbb{R}\mathrm{P}^4)^j \times P(1,2)^k$$

where $i = 0, 1, 2, 3$ and $j = 0, 1$ as well as $k \in \mathbb{N}$.

Good Generators II

Repeatedly applying the above yields, that all (additive) generators are good, except maybe those of form

$$(\mathbb{R}\mathrm{P}^2)^i \times (\mathbb{R}\mathrm{P}^4)^j \times P(1,2)^k$$

where $i = 0, 1, 2, 3$ and $j = 0, 1$ as well as $k \in \mathbb{N}$. But theres more...

Good Generators II

Repeatedly applying the above yields, that all (additive) generators are good, except maybe those of form

$$(\mathbb{R}\mathrm{P}^2)^i \times (\mathbb{R}\mathrm{P}^4)^j \times P(1,2)^k$$

where $i = 0, 1, 2, 3$ and $j = 0, 1$ as well as $k \in \mathbb{N}$. But theres more...

Lemma (Ebert–Wiemeler)

$[P(1,2)]$ is good. Furthermore, elements of form $[M' \times P(1,2)]$ are good, if M' has a PSC-metric.

Good Generators III

There are only finitely many generators left of which we dont know whether they are good.

Good Generators III

There are only finitely many generators left of which we dont know whether they are good.

$$
\begin{aligned}
n = 2, &\quad \mathbb{R}\mathrm{P}^2 \\
n = 4, &\quad \mathbb{R}\mathrm{P}^4, \mathbb{C}\mathrm{P}^2 \\
n = 6, &\quad \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2, \mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^4 \\
n = 8, &\quad \mathbb{R}\mathrm{P}^4 \times \mathbb{C}\mathrm{P}^2 \\
n = 10, &\quad \mathbb{R}\mathrm{P}^2 \times \mathbb{R}\mathrm{P}^4 \times \mathbb{C}\mathrm{P}^2
\end{aligned}
$$

Good Generators III

There are only finitely many generators left of which we dont know whether they are good.

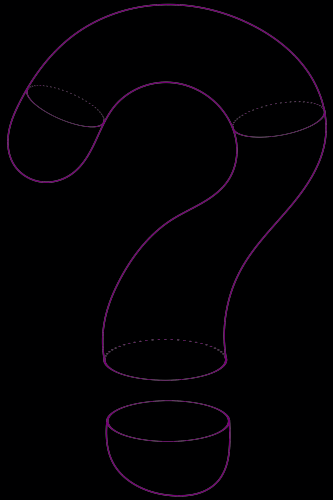
$$
\begin{aligned}
n = 2, &\quad \mathbb{RP}^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 4, &\quad \mathbb{RP}^4, \mathbb{CP}^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 6, &\quad \mathbb{RP}^2 \times \mathbb{CP}^2, \mathbb{RP}^2 \times \mathbb{RP}^4 \\
n = 8, &\quad \mathbb{RP}^4 \times \mathbb{CP}^2 \\
n = 10, &\quad \mathbb{RP}^2 \times \mathbb{RP}^4 \times \mathbb{CP}^2
\end{aligned}
$$

Good Generators III

There are only finitely many generators left of which we dont know whether they are good.

$$
\begin{aligned}
n = 2, &\quad \mathbb{R}P^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 4, &\quad \mathbb{R}P^4, \mathbb{C}P^2 && \leftarrow \text{Theorem doesn't apply} \\
n = 6, &\quad \mathbb{R}P^2 \times \mathbb{C}P^2, \mathbb{R}P^2 \times \mathbb{R}P^4 \\
n = 8, &\quad \mathbb{R}P^4 \times \mathbb{C}P^2 \\
n = 10, &\quad \mathbb{R}P^2 \times \mathbb{R}P^4 \times \mathbb{C}P^2
\end{aligned}
$$

$\rightarrow$ In particular, for dimension > 10 all additive generators are good, and we obtain our theorem.



Main Result

Theorem

Let M be a n-dimensional Manifold of tangential 2-type $\mathrm{id}_{\mathrm{BO}}$, and $n > 10$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^{n-6} \times \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2)$$

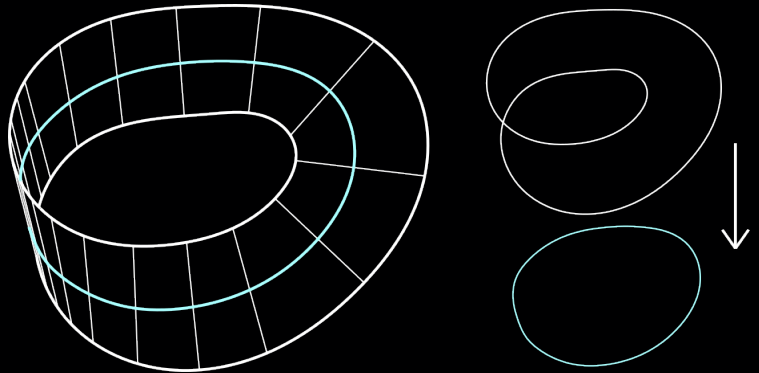
Main Result

Theorem

Let M be a n-dimensional Manifold of tangential 2-type $\mathrm{id}_{\mathrm{BO}}$, and $n > 10$. Then

$$\mathcal{R}^+(M) \simeq \mathcal{R}^+(S^{n-6} \times \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2)$$

$\rightarrow$ The manifold $S^{n-6} \times \mathbb{R}\mathrm{P}^2 \times \mathbb{C}\mathrm{P}^2$ is just an arbitrary reference manifold with the correct tangential 2-type.



θ-Structures and Tangential 2-Types

θ-Structures

Definition

Let $\theta\colon B \to \mathrm{BO}$ be a fibration. A θ-structure on a manifold M is a lift of the stable tangent bundle $\tau\colon M \to \mathrm{BO}$ along θ.

θ-Structures

Definition

Let $\theta\colon B \to \mathrm{BO}$ be a fibration. A θ-structure on a manifold M is a lift of the stable tangent bundle $\tau\colon M \to \mathrm{BO}$ along θ.

$\to$ Generalizes orientations via $\theta\colon \mathrm{BSO} \to \mathrm{BO}$.

θ-Structures

Definition

Let $\theta\colon B \to \mathrm{BO}$ be a fibration. A θ-structure on a manifold M is a lift of the stable tangent bundle $\tau\colon M \to \mathrm{BO}$ along θ.

$\to$ Generalizes orientations via $\theta\colon \mathrm{BSO} \to \mathrm{BO}$.

$\to$ Generalizes spin structures via $\theta\colon \mathrm{BSpin} \to \mathrm{BO}$.

θ-Structures

Definition

Let $\theta\colon B \to \mathrm{BO}$ be a fibration. A θ-structure on a manifold M is a lift of the stable tangent bundle $\tau\colon M \to \mathrm{BO}$ along θ.

$\to$ Generalizes orientations via $\theta\colon \mathrm{BSO} \to \mathrm{BO}$.

$\to$ Generalizes spin structures via $\theta\colon \mathrm{BSpin} \to \mathrm{BO}$.

$\to$ Every manifold has a canonical $\mathrm{id}_{\mathrm{BO}}$-structure.

Tangential Type

Definition

A θ-structure on M is called tangential k-type of M, if the θ is k-coconnected and the lift is k-connected.

Tangential Type

Definition

A θ-structure on M is called tangential k-type of M, if the θ is k-coconnected and the lift is k-connected.

$\rightarrow$ Unique up to fiber homotopy equivalence.

Tangential Type

Definition

A θ-structure on M is called tangential k-type of M, if the θ is k-coconnected and the lift is k-connected.

$\rightarrow$ Unique up to fiber homotopy equivalence.

$\rightarrow$ Tangential 1-type is either $\mathrm{BSO} \to \mathrm{BO}$ or $\mathrm{id}_{\mathrm{BO}}$, depending on w_1.

Tangential 2-types

Theorem

Let $G = \pi_1(M)$ and w_i denote the Stiefel–Whitney-classes of M. Call $\widetilde{w}_2 = w_2(\widetilde{M})$. Then the tangential 2-type of M is:

- $\mathrm{BO} \times_{\mathrm{B}\mathbb{Z}_2} \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 \neq 0$
- $\mathrm{BSO} \times \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 = 0$
- $\mathrm{B}(G, w_1, w_2)$, if $\widetilde{w}_2 = 0$.

Tangential 2-types

Theorem

Let $G = \pi_1(M)$ and w_i denote the Stiefel–Whitney-classes of M. Call $\widetilde{w}_2 = w_2(\widetilde{M})$. Then the tangential 2-type of M is:

- $\mathrm{BO} \times_{\mathrm{B}\mathbb{Z}_2} \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 \neq 0$
- $\mathrm{BSO} \times \mathrm{B}G$, if $\widetilde{w}_2 \neq 0, w_1 = 0$
- $\mathrm{B}(G, w_1, w_2)$, if $\widetilde{w}_2 = 0$.

$\rightarrow$ These are six cases, as $\widetilde{w}_2 \neq 0$ implies $w_2 \neq 0$.